MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8596

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode; ceramic-metal, Fl = 400 MHz

Outline: See figure 1

Mounting position: Any

ABSOLUTE RATINGS:

Parameters: / Unit:	Ef V (See note 1)	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ib mAdc	Ic1 mAdc
AB Audio	6.3 ±10%	1,000	---	300	180	30
AB SSBC	6.3 ±10%	1,000	-100	300	250	30
C Telep	6.3 ±10%	800	-100	300	150	30
C Teleg	6.3 ±10%	1,000	-100	300	180	30
TEST CONDITIONS:	6.3	1,000	Adjust	300	---	--

ABSOLUTE RATINGS:

Parameters: / Unit:	$P(g^2)$ W	Pi W	Pp W	Seal Temp °C	tk sec	Barometric pressure reduced mm Hg
AB Audio	4.5	180	115	250	60	140.7
AB SSBC	4.5	180	115	250	60	140.7
C Telep	3.0	120	75	250	60	140.7
C Teleg	4.5	180	115	250	60	140.7
TEST CONDITIONS: (See note 2)	---	---	---	250 max	--	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	3					
1301	Heater current	---		If	2.90	3.55	A
1266	Total grid current	4	Ec1/Ib = 115 mAdc	Ic1	---	-20	µAdc
1256	Electrode current (screen)	---	Ec1/Ib = 115 mAdc	Ic2	-8.0	+2.0	mAdc
1261	Electrode voltage (1) (grid)	---	Ec1/Ib = 115 mAdc	Ec1	-6.0	-13.5	Vdc
1261	Electrode voltage (2) (grid)	---	Ec1/Ib = 1 mAdc	Ec1	---	-47	Vdc
1231	Pulsing emission	5	etd/ik = 10 a	etd	---	400	v
1372	Current division (anode)	---	Eb = 350 Vdc; Ec1 = -100 Vdc procedure (b)	Ib	390	---	ma
	Quality conformance inspection, part 2	6					
1266	Primary grid emission (control)	---	Ec1/g1 input = 2 W; t = 30; a and g2 grounded	Isg1	---	-12	µAdc
1266	Primary grid emission (screen)	---	Ec2/g2 input = 4.5 W; a and g1 grounded; t = 30	Isg2	---	-25	µAdc
1366	Interelement leakage resistance, cold	7		R	1.0	---	Meg
2214	RF useful power output	8	Ef = 5.7 V; t = 120; class C amplifier, F = 400 ±20 MHz; Ib = 180 mAdc; Ic1 = 30 mAdc (max); Pd = 3.3 W (max)	Po	80	---	W (useful)
1236	Power oscillation		F = 15 MHz; Eb = 850 Vdc; Ec2 = Adjust; Ib = 150 mAdc; Rg1 = 2,000 ohms; Ic1 = 30 mAdc; t = 120	Po	80	---	W (useful)
1331	Direct-interelectrode capacitance	---	Capacitance fixture per 289-JAN; all unused electrodes grounded	Cg1k Cg1g2 Cg2p Cg1p Cpk Cg2k	11.8 17.3 4.0 --- --- ---	15.2 21.9 5.1 .065 .019 1.30	pF pF pF pF pF pF

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3	6					
----	Life test	9	Group C; Ef = 5.7 V; Class C amplifier; F = 400 ±20 MHz; Ib = 180 mAdc; Ic1 = 30 mAdc max; Ec1 = value obtained in Ec1 test per Part 1 herein	t	500	---	hrs
----	Life test end points:						
1231	Pulsing emission	---	etd/ik = 7.5 a	etd	---	400	v
1266	Primary grid emission (control)	---	Ec1/g1 input = 2 W; t = 30; a and g2 grounded	Isg1	---	-12	µAdc
1266	Primary grid emission (screen)	---	Ec2/g2 input = 4.5 W; t = 30; a and g1 grounded	Isg2	---	-25	µAdc
1506	Heater-cycling life	---	Group C; Ef = 8.5 V; t = 150 "ON", 300 "OFF"	---	384	---	cycles
----	Heater-cycling life test end point:						
1301	Heater current	---		If	2.32	4.26	A
1042	Shock, specified pulse (1)	10	Test condition A; Ef only	---	---	---	---
1041	Shock (2)	10	500G, 0.75 ms	---	---	---	---
----	Vibration (1)	10, 11	Eb = 300 Vdc; Ec2 = 250 Vdc; Ec1/Ib = 10 mAdc; Rp = 2,000 ohms	Ep	---	15	Vrms
----	Vibration (2)	10, 12	Eb = 300 Vdc; Ec2 = 250 Vdc; Ec1/Ib = 10 mAdc; Rp = 2,000 ohms Rg2 = 1,000 ohms	---	---	---	---
----	Shock (1), shock (2) and vibration (2) end points:	---					
1266	Total grid current	---	Ec1/Ib = 115 mAdc	Ic1	---	-20	µAdc
1261	Electrode voltage (1) (grid)	---	Ec1/Ib = 115 mAdc	Ec1	-6.0	-18	Vdc
1002	Barometric pressure, reduced	10	Pressure = 140.7 mm Hg; 1,300 Vdc applied to anode with g1 and g2 tied to k	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.

NOTES:

1. The tube under test (TUT) cathode will be subjected to considerable back bombardment as the frequency is increased, due to transit-time effects which raise the temperature of the cathode. Depending upon operating conditions and frequency, the heater voltage should be reduced to prevent overheating of the cathode and resultant short life.

2. Pertinent to the specified test conditions and performance of all electrical tests, a dummy 7457 type radiator may be attached to the anode of type 8596 when under test.

3. The AQL for all tests listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

4. Airflow shall be supplied to limit the TUT anode seal temperature to 225°C minimum.

5. The applied voltage source shall meet the following requirements: Pulse duration = 2 µs minimum and not greater than 10 µs at the base line of the pulse, prr = 60 Hz minimum. Adjust the pulse amplitude until the specified peak cathode current is obtained. Measurement shall be made at end of one minute or when stability is reached, whichever occurs first.

6. Quality Conformance Inspection, Part 2 and Part 3 tests may be performed on any of the types of the same basic 8596 construction manufactured during any one month's production period.

7. Prior to performing this test, the TUT shall be cooled at room ambient temperature for 30 minutes. Using the test circuit shown on figure 1366-1 of Test Method 1366, measure the resistance in both directions, between any two adjacent electrodes, except across the heater terminals. If the resistance value measured is below that required, repeat this test after a 10-minute interval. The TUT shall be rejected if it fails again.

8. This test shall be performed using a grid-driven tuned amplifier. Adjust Ecl bias supply, and tune circuit for maximum useful power output. The specified driver power output shall be measured using a Bird Thru-line Wattmeter, or equivalent. Driving power output = forward power minus reflected power. Ecl supply shall have an effective impedance of 500 ohms maximum.

9. The TUT shall operate in a grid-driven tuned amplifier; drive shall be varied and output circuit tuned for maximum power output with specified Ib; Ecl supply shall have an effective impedance of 500 ohms maximum.

10. This test shall be performed (or dimensions shall be checked) during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consective samples have been accepted.

11. The TUT shall be vibrated in positions X, Y, and Z under the following conditions:

Frequency (Hz)	Double Amplitude (inches)	Acceleration (G)	Approx. Sweep Duration (minutes)
5-10	0.080 ±10%	---	1
10-15	---	0.41 ±10%	.5
15-75	0.036 ±10%	---	.5
75-2000	---	10 ±10%	6

Cycling test: The frequency shall be swept from 5 to 2,000 to 5 Hz with approximately logarithmic progression, and shall require approximately 8 minutes to traverse the range. This constitutes one cycle. The TUT shall be vibrated for one such cycle in each of the three planes.

Record maximum vibrational noise output from 5-75 Hz using the Bruel and Kjaer type 2409 true rms meter, or equivalent. Record vibrational noise output response curve from 75-2000 Hz using the Bruel and Kjaer automatic recorder, or equivalent.

Source: https://assist.dla.mil -- Downloaded: 202640-15T15:02Z
Check the source to verify that this is the current version before use.

NOTES: -Continued

12. The TUT shall be vibrated in positions X, Y and Z under the following conditions:

Frequency (Hz)	Double Amplitude (inches)	Acceleration (G)
5-15	0.170 ±10%	----
15-75	0.030 ±10%	----
75-1500	----	3 ±10%

Cycling test: The frequency shall vary from 5 to 1,500 to 5 Hz with approximately logarithmic progression, and shall require approximately 6 minutes to traverse the range. This consitutes one cycle. The TUT shall be vibrated for one such cycle in each of the three planes.

An oscilloscope having a vertical amplifier with a minimum bandpass of DC to 10 MHz shall be used to monitor the signal across the 2K series plate resistor. The horizontal scope sweep frequency shall be approximately 1 centimeter per second. The signal shall be DC coupled to the oscilloscope. If a signal greater than 150 volts occurs during the vibration sweep, the sweep shall be reversed (first rescan) as soon as possible to pass through and below the frequency where this signal was noted. At this point the sweep shall be reversed to the normal direction (second rescan). If a signal greater than 150 volts recurs (within approximately ±20 percent of the noted frequency) during the first or second rescans the tube shall be considered a failure. If no signal greater than 150 volts occurs during either of the two rescans, the tube is not a failure and the normal vibration sweep shall be continued. If additional signals occur greater than 150 volts, the above procedure shall be repeated.

A signal greater than 25 volts for a duration greater than 2 seconds is cause for rejection.

The 150 volts and the 25 volts referred to above are deflection only and does not include the oscilloscope signal resulting from the 10 ma plate current flowing through Rp.

(NOTE: An equivalent method for measuring the signal across the plate resistor may be used.)

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 85

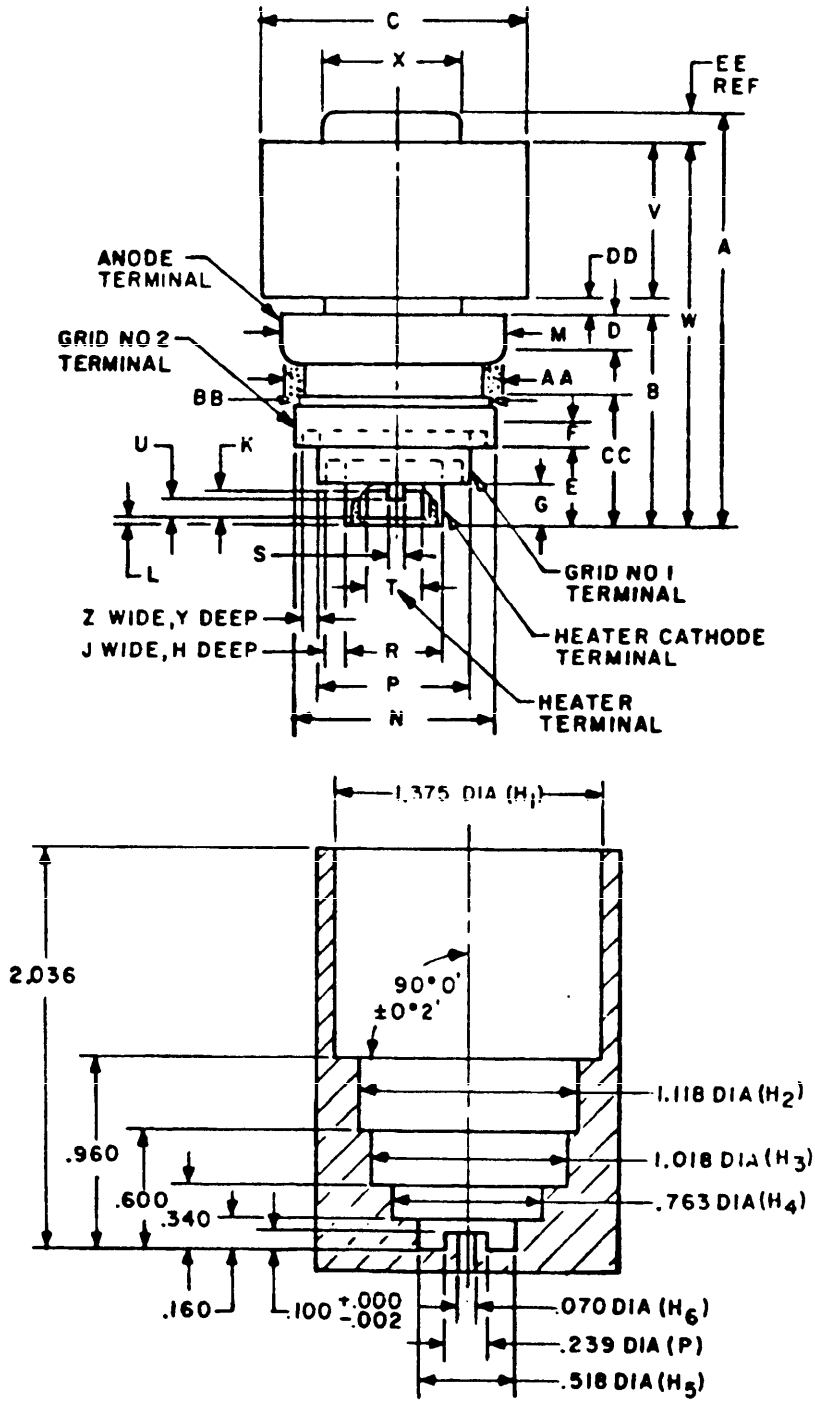
Review activity:
Air Force - 99

User activity:
Air Force - 11

Preparing activity:
Air Force - 85

(Project 5960-F175)

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:02Z
Check the source to verify that this is the currentversion before use.



GAUGE NO. 4

ACTUAL TUBE OUTLINE WITHIN THESE AREAS IS NOT CONTROLLED BY DIMENSIONS AND MUST NOT BE USED FOR ANY PURPOSE

FIGURE 1. Outline drawing of electron tube type 8596.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.

Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
Quality conformance inspect., part 2					
A	1.894	2.036	48.11	51.71	
B	1.000	1.060	25.40	26.92	
C	1.297	1.327	32.94	33.71	
E	.350	.390	8.89	9.91	
G	.160	.190	4.06	4.83	
M	1.085	---	27.56	---	
N	.985	---	25.02	---	
P	.735	---	18.67	---	
R	.480	---	12.19	---	
T	.240	.260	6.10	6.60	
V	.735	.765	18.67	19.43	
W	1.785	1.895	45.34	48.13	
Quality conformance inspect., part 3 (see note 7)					
D	.165	---	4.19	---	
F	.140	---	3.56	---	
H	.120	---	3.05	---	
J	.095	---	2.41	---	
K	.100	---	2.54	---	
L	0	.050	0	1.27	
U	.054	---	1.37	---	
Y	.060	---	1.52	---	
Z	.090	---	2.29	---	
CC	.600	---	15.24	---	
DD	.050	---	1.27	---	
Reference dimensions					
S	---	---	---	---	
X	.690		17.53	nominal	
AA	---	---	---	---	
BB	---	---	---	---	
EE	.125		3.18	nominal	

INCHES	MM	INCHES	MM
.000	.00	.600	15.24
.002	.05	.763	19.38
.070	1.78	.960	24.38
.100	2.54	1.018	25.86
.160	4.06	1.118	28.40
.239	6.07	1.375	34.93
.340	8.64	2.036	51.71
.518	13.16		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.001 (.03 mm).

FIGURE 1. Outline drawing of electron tube type 8596 - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-15T15:02Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Directorate of Electronic Support
Attn: AFALD/PTS
Gentile AFS OH 45444

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-E-1/1587C(USAF)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:02Z
Check the source to verify that this is the current version before use.